

$$1 > \frac{p_e}{p_o} = \left(1 + \frac{\gamma+1}{2} M_e^2\right)^{-\frac{\gamma}{\gamma-1}} \text{ for convergent nozzle } M_e=1 \Rightarrow \frac{p_e}{p_o} = \left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma}{\gamma-1}} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$$

If the nozzle does not have a diverging portion, then the pressure ratio p_e/p_o is given by Eq. (3.11) with $M = 1$:

$$p_e/p_o = \left(\frac{2}{\gamma+1}\right)^{\gamma/(\gamma-1)}$$



$$C_T = \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}} \left[1 - \left(\frac{p_e}{p_o}\right)^{\frac{\gamma-1}{\gamma}}\right]} + \frac{p_e - p_o}{p_o} \frac{A_e}{A^*} \quad (11.11)$$

and the thrust coefficient reduces to

$$C_{T_{\text{conv}}} = (\gamma+1) \left(\frac{2}{\gamma+1}\right)^{\gamma/(\gamma-1)} - \frac{p_a}{p_o}$$

Verify 11.12

3rd proof (11.12)

in which $C_{T_{\text{conv}}}$ is the thrust coefficient for a converging nozzle. The ratio of thrusts, with and without a diverging section, is given by

$$\frac{T}{T_{\text{conv}}} = \frac{C_T}{C_{T_{\text{conv}}}} \stackrel{11.11}{=} \stackrel{11.12}{=}$$

Put $\frac{p_e}{p_o}$ in 11.11
get 11.12

and may be evaluated by means of Eqs. (11.11) and (11.12) above as a function of γ , p_e/p_o , and p_a/p_o . For the special case $\gamma = 1.2$, Fig. 11.3 shows how the diverging portion of the nozzle contributes to the rocket thrust at various area and pressure ratios. We can see that for each value of the ambient pressure ratio p_a/p_o there is an optimum area ratio for the nozzle; it is the one for which $p_e = p_a$ (see discussion of Fig. 10.2).

We can see from Fig. 11.3 that the thrust of a rocket will actually be reduced by adding a diverging section of too large an area ratio. However for reasonable pressure ratios ($p_a/p_o < 0.02$) this would not happen unless A_e/A^* were greater than about 30. For very small pressure ratios (such as $p_a/p_o = 0.001$), we will see that very large area ratios are associated with maximum thrust.

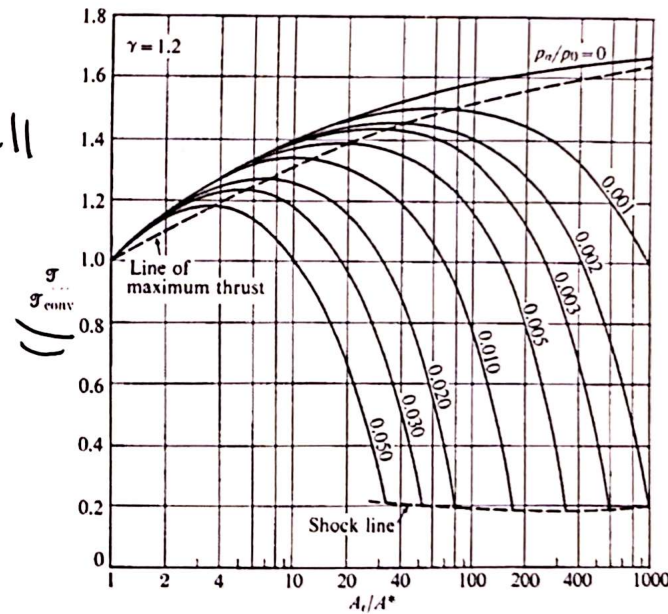


FIGURE 11.3 Performance characteristics of a one-dimensional isentropic rocket nozzle; $\gamma = 1.2$. (After Malina [1].)

Solve $\left(\frac{A_e}{A^*}\right) Eq 7$, get Me , put in $\frac{P_e}{P_o} = \left(1 + \frac{\gamma-1}{2} Me^2\right)^{-\frac{\gamma}{\gamma-1}}$ → get $\frac{P_e}{P_o}$

